



**Bharati Vidyapeeth's
College of Engineering, New Delhi**

**celebrating
Golden Jubilee**
BHARATI VIDYAPEETH
Hon'ble Founder Dr. Patangrao Kadam

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A stylized logo for the Golden Jubilee, featuring the number '50' in a blue and orange color scheme.

FUELS, Lecture-1

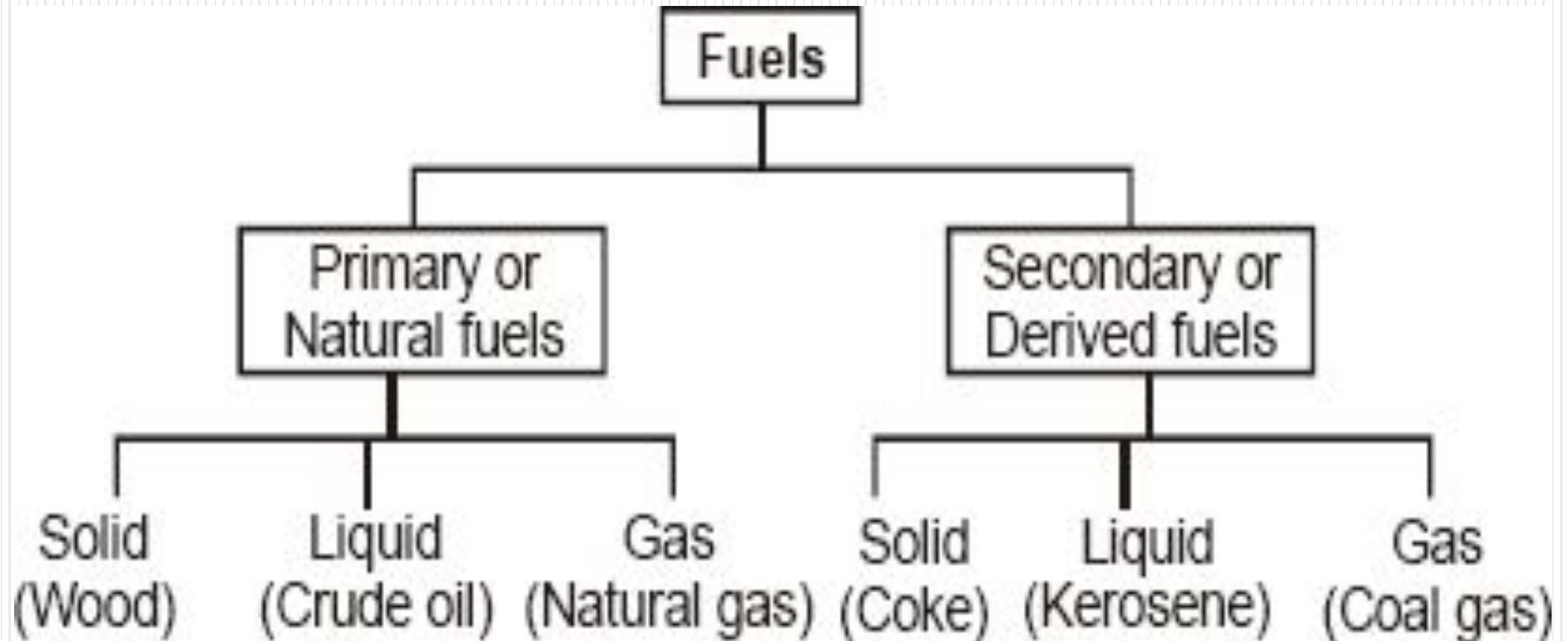
INTRODUCTION

- Fuel is a combustible substance, containing carbon as main constituent, which on proper burning gives large amount of heat, which can be used economically for domestic and industrial purposes. Eg., Wood, Charcoal, Coal, Kerosene, Petrol, Producer gas, Oil gas, LPG etc.,
During the process of combustion of a fuel (like coal), the atoms of carbon, hydrogen, etc. combine with oxygen with the simultaneous liberation of heat at a rapid rate.
- Fuel + Oxygen $\rightarrow$ Products + Heat

CLASSIFICATION OF FUELS:

- Chemical Fuels: It is of two types viz.,
- Primary or Natural Fuel and
- Secondary or Derived Fuel.

Classification of Fuel



CALORIFIC VALUE

- Calorific value of a fuel is “the total quantity of heat liberated, when a unit mass (or volume) of the fuel is burnt completely”

Net calorific value $\approx$ Gross calorific value – 10%



Where:

C = Carbon

H = Hydrogen

O = Oxygen

N = Nitrogen

Units of Heat:

- (1) Calorie- is the amount of heat required to raise the temperature of one gram of water through one degree centigrade (15-16° C).
- (2) Kilocalorie – is equal to 1,000 calories. This is the unit of metric system and may be defined as “the quantity of heat required to raise the temperature of one kilogram of water through one degree centigrade. Thus, 1 kcal = 1,000 calories.
- (3) British Thermal Unit (BTU)- is defined as “the quantity of heat required to raise the temperature of one pound of water through one degree Fahrenheit (60-61° F). This is the English system unit.
 $1 \text{ BTU} = 252 \text{ cal} = 0.252 \text{ kcal}$ and $1 \text{ kcal} = 3.968 \text{ BTU}$
- (4) Centigrade heat unit (CHU)-is “the quantity of heat required to raise the temperature of 1 pound of water through one degree centigrade”. Thus,
 $1 \text{ kcal} = 3.968 \text{ BTU} = 2.2 \text{ CHU}$

HIGHER OR GROSS CALORIFIC VALUE:

- It is the total amount of heat produced, when unit mass/volume of the fuel has been burnt completely and the products of combustion have been cooled to room temperature (15°C or 60°F).
- It is explained that all fuels contain some hydrogen and when the calorific value of hydrogen containing fuel is determined experimentally, the hydrogen is converted into steam. If the products of combustion are condensed to the room temperature, the latent heat of condensation of steam also gets included in the measured heat which is then called GCV.

LOWER OR NET CALORIFIC VALUE

- It is the net heat produced, when unit mass/volume of the fuel is burnt completely and the products are permitted to escape.
- In actual practice of any fuel, the water vapour and moisture, etc., are not condensed and escape as such along with hot combustion gases. Hence, a lesser amount of heat is available.

Lower calorific value of the fuel = HCV – Latent heat of water liberated by the fuel

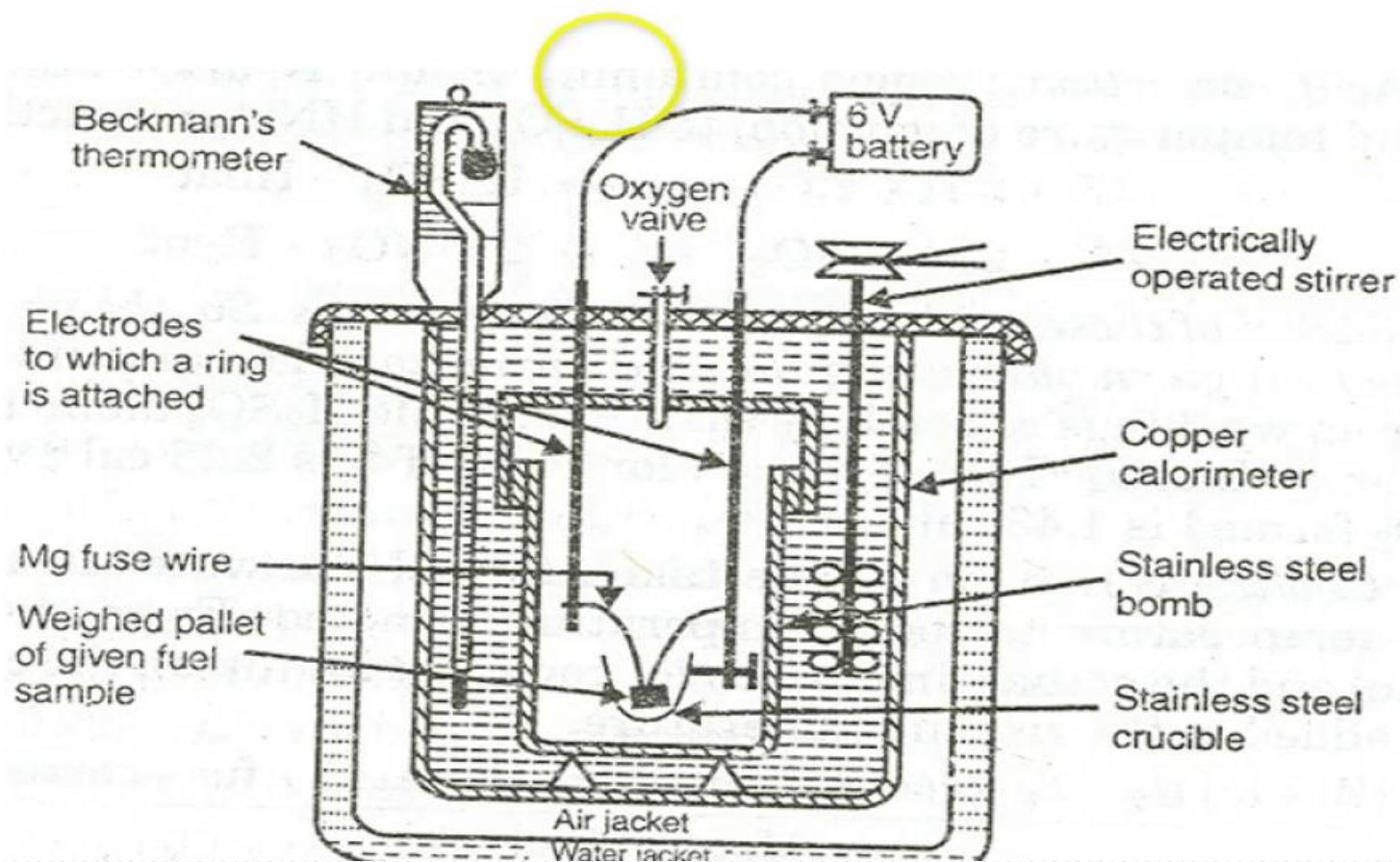
Importance of Calorific Value

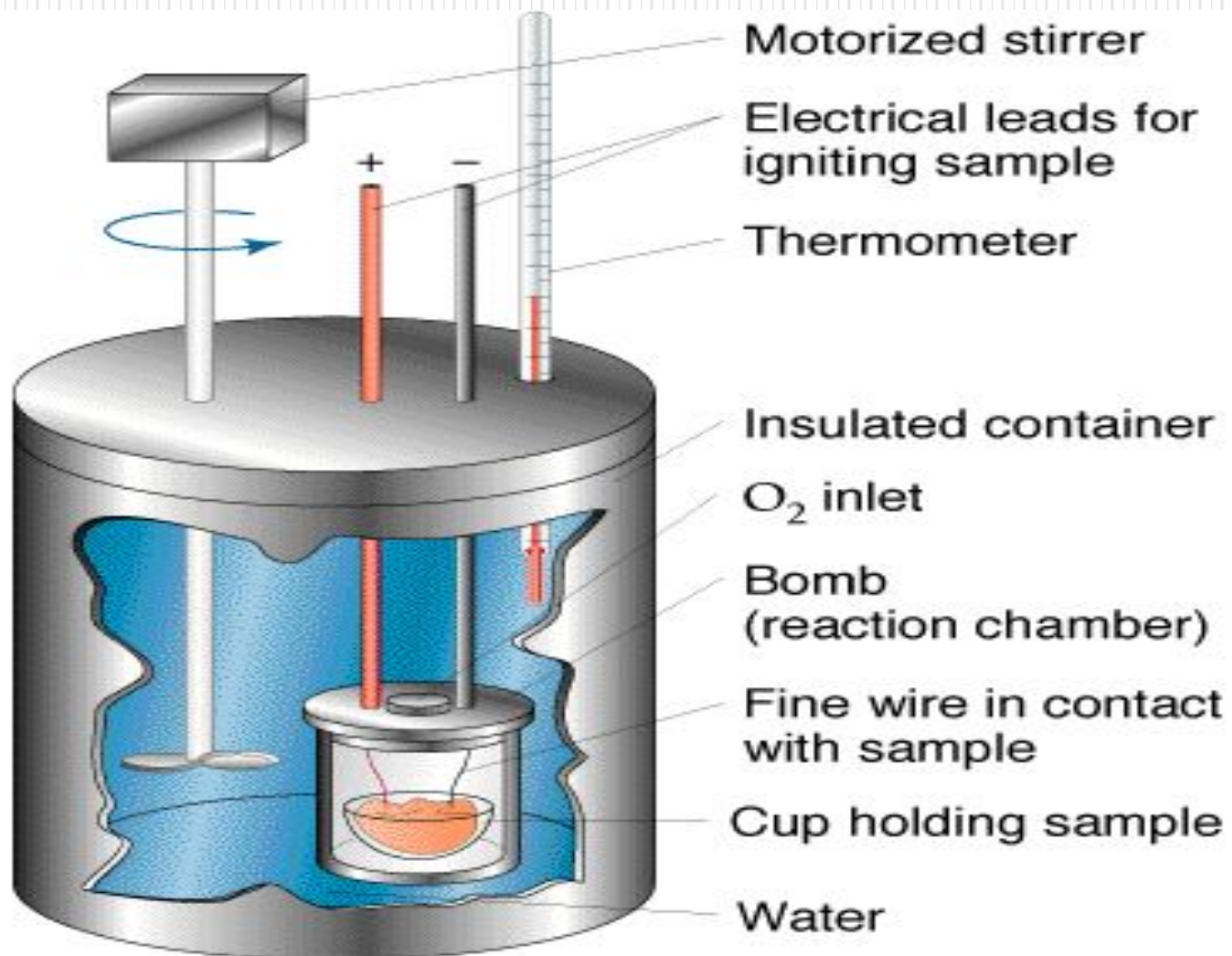
- It is very important to have a knowledge of the calorific value of fuel to carry out our day-to-day activities. This knowledge helps us to determine the amount of energy we transport. The gas shippers and suppliers require this information to bill gas consumers. It also helps to determine transportation charges of gas shippers and suppliers.

DETERMINATION OF CALORIFIC VALUE USING BOMB CALORIMETER

- The calorific value of solid fuels can be determined with the help of bomb calorimeter.
- **Description:**
- Bomb Calorimeter consists of a strong stainless steel bomb where the fuel sample is burnt. The bomb has oxygen inlet valve and two stainless steel electrodes. A small ring is attached to one of the electrodes. In this ring, a nickel or stainless steel crucible is placed. The bomb is placed in a copper calorimeter containing a known weight of water sample. The copper calorimeter is provided with a Beckmann's thermometer and stirrer for stirring water. The copper calorimeter is covered by an air jacket and water jacket.

Bomb Calorimeter Diagram





Functioning:

- A known weight of the fuel sample is taken into the crucible. The fine magnesium wire is touching the fuel sample and then stretched across the electrodes. The bomb lid is tightly closed with the help of screw. The bomb is filled with oxygen at 25 atmospheric pressure.
- The bomb is now placed in a copper calorimeter which containing known weight of water. Initial temperature of the water in the calorimeter is noted ($t_1^{\circ}\text{C}$) after stirring. The electrodes are connected to a battery (6 v). The current is now supplied to the fuel sample which undergoes burning with the evolution of heat. The liberated heat increases the temperature of water in the calorimeter. The maximum temperature of the water during experiment is finally noted ($t_2^{\circ}\text{C}$). From the temperature difference, calorific value of the fuel can be calculated as follows:

Calculation:

- Weight of the fuel sample taken in the crucible = x g
- Weight of water taken in the calorimeter = W g
- Weight of calorimeter and stirrer in terms of water Equivalent = A g
- Initial temperature of water in the calorimeter = $t_1^\circ\text{C}$
- Final temperature of water in the calorimeter = $t_2^\circ\text{C}$
- Heat absorbed by the water = $W (t_2 - t_1)$ cal ----(1)
- Heat absorbed the calorimeter = $A (t_2 - t_1)$ cal ----(2)
- Total heat absorbed by the water = $W(t_2 - t_1) + A(t_2 - t_1)$ cal
- = $(W + A) (t_2 - t_1)$ cal ----(3)
- The relationship between heat liberated by the fuel and HCV is as follows:
- Heat liberated by the fuel = $x \times (\text{HCV})$ ----(4)

Calculation of Higher Calorific Value (HCV):

- Therefore, heat liberated by the fuel = Heat absorbed by the water and calorimeter

X Weight of fuel

- Compare equation (3) and (4), we get
- $x \times (\text{HCV}) = (W+A) (t_2-t_1)$
- **$\text{HCV} = (W+A) (t_2-t_1) / X \text{ cal/g}$**

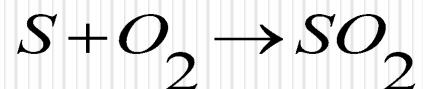
Calculation of Lower Calorific Value (LCV):

- The percentage of hydrogen in the fuel = H
- Weight of water produced 1 g of the fuel = $9 H \text{ g}/100 = 0.09 H \text{ g}$
- Therefore, heat liberated during the Condensation of steam = $0.09 H \times 587 \text{ cal/g}$
- Lower calorific value of the fuel = HCV – Latent heat of water
liberated by the fuel
- **$\text{LCV} = \text{HCV} - (0.09 H \times 587) \text{ cal/g.}$**

Corrections Of Bomb Calorimeter

a. Fuse wire correction: As Mg wire is used for ignition, the heat generated by burning of Mg wire is also included in the gross calorific value. Hence this amount of heat has to be subtracted from the total value.

b. Acid Correction: During combustion, sulphur and nitrogen present in the fuel are oxidized to their corresponding acids under high pressure and temperature.



$$\Delta H = -144,000 \text{ cal}$$

$$\Delta H = -57,160,000 \text{ cal}$$

The corrections must be made for the heat liberated in the bomb by the formation of H_2SO_4 and HNO_3 . The amount of H_2SO_4 and HNO_3 is analyzed by washings of the calorimeter.

For each ml of 0.1 N H_2SO_4 formed, 3.6 calories should be subtracted.

For each ml of 0.1 HNO_3 formed, 1.429 calories must be subtracted.

(C) Cooling correction: As the temperature rises above the room temperature, the loss of heat does occur due to radiation, and the highest temperature recorded will be slightly less than that obtained. A temperature correction is therefore necessary to get the correct rise in temperature.

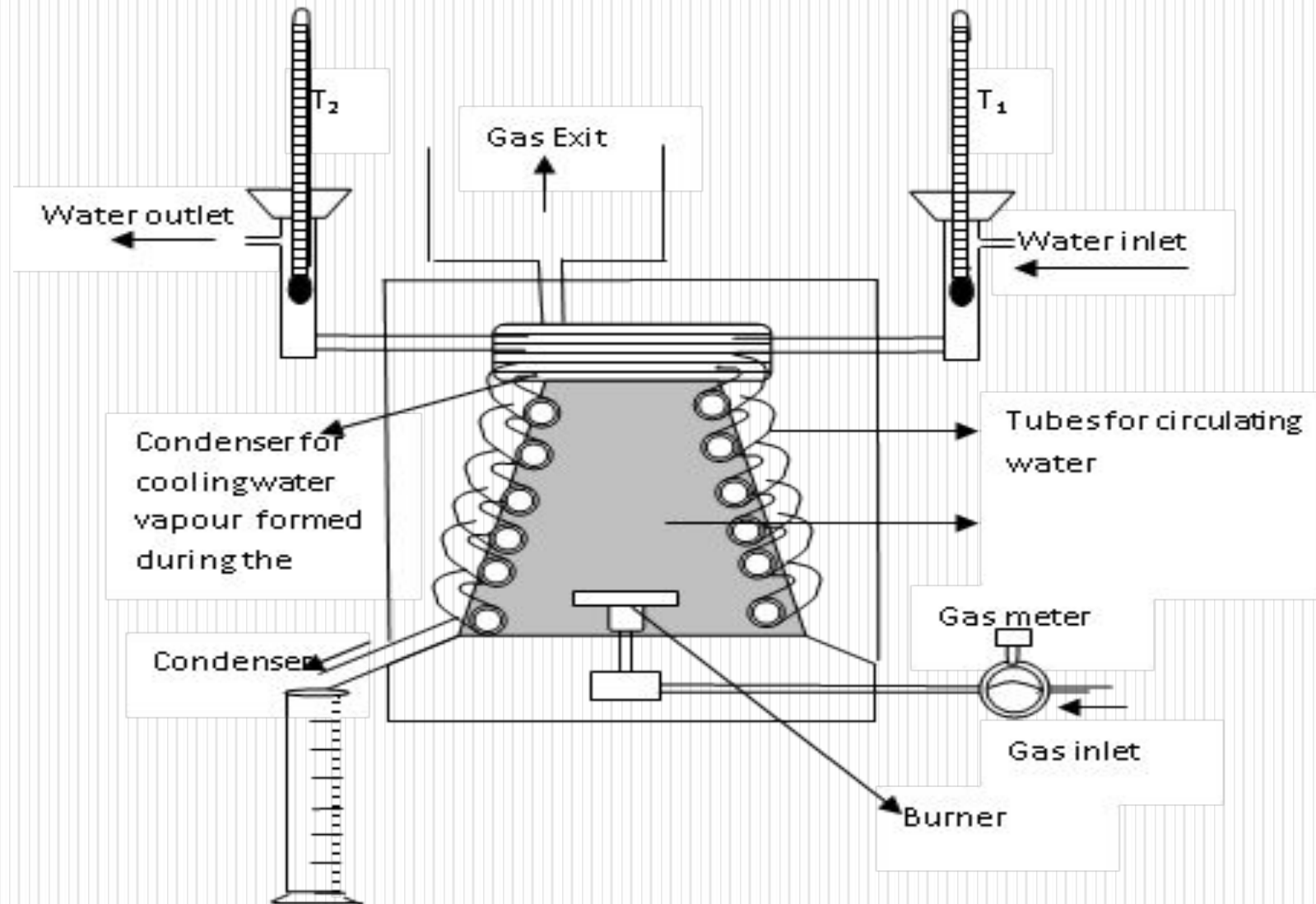
If the time taken for the water in the calorimeter to cool down from the maximum temperature attained, to the room temperature is x minutes and the rate of cooling is dt/min , then the cooling correction = $x \times dt$. This should be added to the observed rise in temperature

Therefore, **Gross calorific value**

$$C = \frac{(W+w)(t_2-t_1) + \text{cooling correction} - [\text{Acid} + \text{fuse correction}]}{\text{Mass of Fuel}}$$

BOYS OR JUNKERS GAS CALORIMETER

- APPARATUS: It consists of following
 - (i) **Bunsen Burner:** special type of Burner clamped at the bottom. It can be pulled out of the combustion chamber or pushed up in chamber during the carrying out combustion.
 - (ii) **Gasometer:** It is employed to measure the volume of gas burning per unit time. This attached with manometer fitted with the thermometer so that pressure and temperature of the gas before burning can be read.
 - (iii) **Pressure governor:** It can control the supply of quantity of gas at give pressure.
 - (iv) **Gas Calorimeter/ Combustion chamber:** It is a vertical cylinder, which is surrounded by annular space for heating water and interchange coils. The entire is covered by an outer jacket in order to reduce the heat loss by radiation and convection.





Functioning:

- Install the equipment on a flat rigid platform near an uninterrupted continuous water source of ½" size and a drain pipe.
- Connect the gas source to the pressure regulator, gas flow meter and the burner respectively in series.
- Insert the thermometer / temperature sensors, into their respective places to measure water inlet and outlet temperatures and a thermometer to measure the flue gas temperature at the flue gas outlet
- Start the water flow through the calorimeter at a study constant flow rate and allow it to drain through over flow.
- Start the gas flow slowly and light the burner out side the calorimeter

When the steady condition are established, then the reading are taken simultaneously of:

- The volume of gaseous fuel burnt (V) at given temperature and pressure in certain period of time (t).
- The quantity of water (W Kg) passing through the annular space during the same interval of time
- The steady rise in temperature ($T_2 - T_1$)
- The mass of water (steam) condensed (in Kg) in the outlet water.
- Calculation of GCV is based on
- Heat lost by Fuel = heat gained by Water

Calculation:

- Volume of gas burn at STP in certain time (t) = V
- Mass of the cooling water used in time $t = W$
- Temperature of inlet water = T_1
- Temperature of outlet water = T_2
- Mass of steam condensed in time t in graduated cylinder = m
- Higher calorific value of fuel = $L = ?$
- Specific heat of water = S
- Heat absorbed by circulating water = $W(T_2 - T_1) \times \text{Specific heat of water (s)}$
- Heat produced by combustion of fuel = VL

Calculation of HCV & NCV:

- Therefore, heat liberated by the fuel = Heat absorbed by the water and calorimeter

$$VL = W(T_2 - T_1) \times S$$

$$HCV (L) = W(T_2 - T_1) \times S / V$$

$$LCV = \left[L - \frac{\text{amount of water collected} \times \text{latent heat}}{V} \right]$$

$$LCV = \left[L - \frac{m \times 587}{V} \right] \text{Kcal/m}^3$$

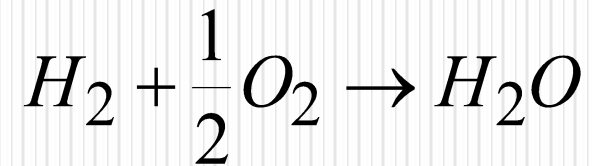
$$1 \text{ cm}^3 \text{ of water} = 1 \text{ g of water}$$

Theoretical calculation of Calorific value of a Fuel

- The calorific value of a fuel can be calculated if the percentages of the constituent elements are known.

● Substrate	Calorific value
● Carbon	8080
● Hydrogen	34500
● Sulphur	2240

- If oxygen is also present, it combines with hydrogen to form H_2O . Thus the hydrogen in the combined form is not available for combustion and is called fixed hydrogen.
- Amount of hydrogen available for combustion = Total mass of hydrogen-hydrogen combined with oxygen.



1g 8g 9g

- Fixed Hydrogen = Mass of oxygen in the fuel
- Therefore, mass of hydrogen available for combustion = Total mass of hydrogen-1/8 mass of oxygen in fuel
- $= H - O/8$

- . Dulong's formula for calculating the calorific value is given as:

Gross calorific Value (HCV)

$$= \frac{1}{100} [8080C + 34,500(H - \frac{O}{8}) + 2,240S] kcal / kg$$

Net Calorific value (LCV)

$$= [HCV - \frac{9H}{100} \times 587] kcal / kg$$

$$= [HCV - 0.09H \times 587] kcal / kg$$

Difference between gross and net calorific value

- **Gross calorific value (GCV)** is the amount of heat released by the complete combustion of a unit of natural gas. It is also known as Higher Heating Value (HCV).
- $GCV = NCV$ when amount of hydrogen will be zero
- **Net Calorific Value (NCV)** also known as lower heating value (LHV) or lower calorific value (LCV) is determined by the subtracting the heat of vaporization of the water vapour from the higher heating value. This treats any H_2O formed as a vapor.
- $NCV = GCV$ when amount of hydrogen will be zero

CHARACTERISTICS OF A GOOD FUEL:

- High calorific value
- Moderate ignition temperature
- Low moisture content
- Low non-combustible matter content
- Moderate velocity of combustion
- Products of combustion should not be harmful
- Low cost
- Easy to transport
- Combustion should be easily controllable
- Should not undergo spontaneous combustion
- Storage cost in bulk should be low
- Should burn in air with efficiency without much smoke
- In case of solid fuel, the size should be uniform so that combustion is regular.

COAL:

- Coal is a highly carbonaceous matter that has been formed as a result of alteration of vegetable matter (eg., plants) under certain favourable conditions. It is chiefly composed of C, H, N and O besides non-combustible inorganic matter.
- The successive stages in the transformation of vegetable matter into coal are – wood, peat, lignite, bituminous coal, steam coal and anthracite. Anthracite is probably the purest form of coal and contains 95 % carbon.

ANALYSIS OF COAL:

- The composition of coal varies widely and hence it is necessary to analyse and interpret the results from the points of view of commercial classification, price fixation and proper industrial utilization. The quality of a coal is ascertained by the following two types of analysis.
- Proximate Analysis and
- Ultimate Analysis

PROXIMATE ANALYSIS:

- It is called 'proximate' because the data collected vary with the procedure adopted.
- It is an empirical but important analysis dealing with the determination of
- Moisture content
- Volatile matter
- Ash content
- Fixed carbon.

(i) Determination of moisture content in coal:

- It is the loss in weight of coal caused by heating a weighed quantity of coal sample for one hour at 105 C.
- **% of moisture content = Loss in weight of coal x 100/ Weight of coal taken initially**
- **Significance of Moisture:**
 - High moisture content of the coal is undesirable for the following reasons:
 - 1) Reduces the calorific value of coal
 - 2) Increases the consumption of coal for heating purpose.
 - 3) Lengthens the time of heating.

(ii) Determination of volatile matter in coal:

- It is the loss in weight of moisture free powdered coal when heated in a
- crucible fitted with cover in a muffle furnace at 950 C for seven minutes.
- **% of volatile matter in coal = Loss in weight of moisture free coal x100/Weight of coal taken**
- **Significance of volatile matter:**
- During burning of coal, gases like CO, CO₂, CH₄, N₂, O₂, hydrocarbons etc. that
- come out are called volatile matter of the coal.
- It has been found that the coal with higher volatile matter content ignites easily has
- lower calorific value burns with long yellow smoky flame will evolve more coal gas when
- heated in the absence of air.

(iii) Determination of ash in coal:

- It is the weight of residue obtained after burning a known weight of coal in an
- open crucible in the presence of air at 750 C for 30 minutes.
- **% of ash in coal = Weight of residue ash formed x 100/Weight of coal taken**
- **Significance of ash content:**
- 1) Ash is the combination product of mineral matter in the coal. It consists mainly
- SiO_2 , Al_2O_3 and Fe_2O_3 with varying amount of other oxides such as Na_2O , CaO , MgO etc.
- 2) Ash containing the oxides of Na, Ca and Mg melt easily.
- 3) High ash content in coal is undesirable because it
- i) increases transporting, handling and storage costs
- ii) is harder and stronger
- iii) has lower calorific value

(iv) Determination of fixed carbon:

- It is determined indirectly by deducting the sum total of moisture, volatile matter and
- ash percentage from 100.
- **% of fixed carbon in coal = 100 - % (moisture + volatile matter + ash)**
- **Significance of Fixed carbon:**
- It is the pure carbon present in coal. Higher the fixed carbon content of the coal,
- higher will be the calorific value of the sample.
- The proximate analysis data of some Indian coal samples are presented here.

The proximate analysis data of some Indian coal samples are presented here.

N o.	Source of sample	Moisture (%)	VM (%) Ash (%)	Ash (%)	Fixed carbon (%)
1	Assam	1.2 – 1.5	35-38	1.6-2	55-60
2	Jharia	0.6-2.0	15-25	10-20	60-68
3	Raniganj	4-6	30-35	9-15	50-60

ULTIMATE ANALYSIS

- Ultimate analysis refers the determination of weight percentage of carbon, hydrogen, nitrogen, oxygen and sulphur of pure, dry coal.
- This analysis gives the elementary, ultimate constituents of coal.
- This analysis is essential for calculating heat balances in any process for which coal is employed as a fuel.
- It is useful to the designing of coal burning equipments and auxiliaries.

i) Determination of carbon and hydrogen in coal:

- A known amount of coal is burnt in presence of oxygen thereby converting carbon and hydrogen of coal into-
- (i) CO_2 ($\text{C} + \text{O}_2 \rightarrow \text{CO}_2$) and
- (ii) H_2O ($\text{H}_2 + \frac{1}{2} \text{O}_2 \rightarrow \text{H}_2\text{O}$) respectively.
- The products of combustion CO_2 and H_2O are passing over weighed tubes of anhydrous CaCl_2 and KOH which absorb H_2O and CO_2 respectively.
- The increase in the weight of CaCl_2 tube represents the weight of water formed while the increase in the weight of KOH tube represents the weight of CO_2 formed.

The percentage of carbon and hydrogen in coal can be calculated in the following way-

- The weight of coal sample taken = x g
- The increase in the weight of KOH tube = y g
- The increase in the weight of CaCl₂ tube = z g
- Consider the following reaction
- $\text{C (12)} + \text{O}_2 \longrightarrow \text{CO}_2(44)$
- 44 g of CO₂ contains 12 g of carbon
- Therefore y g of CO₂ contains $\frac{y \times 12}{44}$ g of carbon.
- X g of coal contains $\frac{12 y}{44}$ g carbon
- % of carbon in coal $= \frac{12 y}{44x} \times 100$

Significance of Total Carbon:

- It is the sum total of fixed carbon and the carbon present in the volatile matters like CO, CO₂, hydrocarbons. Thus, total carbon is always more than fixed carbon in any coal. High total carbon containing coal will have higher calorific value.

(ii) Determination of hydrogen:

- Consider the following reaction.
- $\text{H}_2 (2) + \frac{1}{2} \text{O}_2 \longrightarrow \text{H}_2\text{O}(18)$
- 18 g of water contains 2 g of hydrogen.
- Z g of water contains $\frac{2 z}{18}$ of hydrogen
- X g of coal contains $\frac{2 z}{18}$ g of hydrogen
- % of hydrogen in coal $= \frac{2 z}{18} \times 100$

Significance of Hydrogen:

- It increases the calorific value of the coal. It is associated with the volatile matter of the coal. When the coal containing more of hydrogen is heated, it combines with nitrogen present in coal forming ammonia. Ammonia is usually recovered as $(\text{NH}_4)_2\text{SO}_4$, a valuable fertilizer.

(ii) Determination of nitrogen:

- This is done by Kjeldhal's method.
- A known amount of powdered coal is heated with concentrated sulphuric acid in the presence of K_2SO_4 and $CuSO_4$ in a long necked Kjeldhal's flask. This converts nitrogen of coal to ammonium sulphate. When the clear solution is obtained (ie., the whole of nitrogen is converted into ammonium sulphate), it is heated with 50 % NaOH solution and the following reaction occurs:
- $(NH_4)_2SO_4 + 2 NaOH \rightarrow Na_2SO_4 + 2 NH_3 + H_2O$

Cont.....

- The ammonia thus formed is distilled over and is absorbed in a known quantity of standard 0.1 N H₂SO₄ solution. The volume of unused 0.1 N H₂SO₄ is then determined by titrating against standard NaOH solution. Thus, the amount of acid neutralized by liberated ammonia from coal is determined using the formula.
- $$\% \text{ Nitrogen in coal} = \frac{14 \times \text{volume of acid used} \times \text{normality of acid} \times 100}{1000 \times \text{wt of fuel sample taken}}$$

Or
$$= \frac{1.4 \times \text{volume of acid used} \times \text{normality of acid}}{\text{Wt. of fuel sample taken}}$$

Significance:

Presence of nitrogen decreases the calorific value of the coal. However, when coal is carbonized, its N_2 and H_2 combine and form NH_3 . Ammonia is recovered as $(NH_4)_2SO_4$, a valuable fertilizer.

(iv) Determination of sulphur in coal:

A known amount of coal is burnt completely in Bomb calorimeter in presence of oxygen.

Ash thus obtained contains sulphur of coal as sulphate which is extracted with dil.HCl.

The acid extract is then treated with BaCl_2 solution to precipitate sulphate as BaSO_4 .

The precipitate is filtered, washed, dried and weighed. From the weight of BaSO_4 , the percentage of sulphur in coal is calculated in the following way.

The weight of coal sample taken = x g

The weight of BaSO_4 precipitate = y g

Cont.....

consider the following equations



233 g of BaSO₄ contains 32 g of sulphur.

Therefore, y g of BaSO₄ contains = $\frac{32}{233} y$ g sulphur.

Therefore x g of coal contains = $\frac{32}{233} y$ g sulphur.

% of sulphur in the coal = $\frac{32}{233} y \times 100$.

Significance:

It increases the calorific value of the coal, yet it has the following undesirable effect-

The oxidation products of sulphur (SO_2 , SO_3) especially in presence of moisture forms sulphuric acid which corrodes the equipment and pollutes the atmosphere.

v) Determination of oxygen in coal:

It is calculated indirectly in the following way-

$\% \text{ of oxygen in coal} = 100 - \% (\text{C} + \text{H} + \text{N} + \text{S} + \text{ash}).$

Significance:

The less the oxygen content, the better is the coal. As the oxygen content increases, its moisture holding capacity also increases.